

MongoDB Setup and CRUD Operations Guide

1. Creating a MongoDB Atlas Cluster

1. Visit MongoDB Atlas and log in.
2. Click **Create a New Cluster** → choose the **Free M0 Cluster**.
3. Select a cloud provider and region → click **Create Cluster**.
4. Wait a few minutes for setup completion.

2. Creating a Database User

- Go to **Database Access** → **Add New Database User**.
- Set a username and password (keep them secure).
- Assign role **Read and write to any database** → click **Add User**.

3. Network Access Configuration

- Go to **Network Access** → **Add IP Address**.
- Choose **Allow access from anywhere** or specify your IP.
- Click **Confirm**.

4. Retrieving the Connection String

- Go to **Clusters** → **Connect** → **Connect your application**.
- Copy the connection string:
`mongodb+srv://<username>:<password>@cluster0.mongodb.net/<dbname>?retryWrites=true&w=majority`
- Replace placeholders with your actual details.

5. Connecting via MongoDB Compass

1. Download and install MongoDB Compass.
2. Paste your connection string and replace **<password>**.
3. Click **Connect** to access your database visually.

6. Connecting MongoDB with Node.js

Install the MongoDB driver:

```
npm install mongodb
```

Example code:

```
const { MongoClient, ObjectId } = require('mongodb');  
const url = 'mongodb://localhost:27017';  
const client = new MongoClient(url);
```

```
const dbName = 'Namaste-Nodejs';

async function main() {
  await client.connect();
  console.log('Connected successfully to MongoDB');
  const db = client.db(dbName);
  const collection = db.collection('User');
  return 'done';
}
main().then(console.log).catch(console.error).finally(() =>
client.close());
```

7. CRUD Operations Overview

Create: Add new documents to the collection.

Read: Retrieve documents from the collection.

Update: Modify existing documents.

Delete: Remove documents permanently.

CRUD Example Code:

```
// Create
const data = { firstname: "Akshad", lastname: "Jaiswal", city:
"Pune" };
await collection.insertMany([data]);

// Read
const findData = await collection.find({}).toArray();

// Update
await collection.updateOne({ _id: new
ObjectId('67066d6a3be8f41630d5dae4') }, { $set: { firstname:
"Mint" } });

// Delete
await collection.deleteOne({ _id: new
ObjectId('670668562c6bd11e25050c13') });
```

8. Summary

By following this guide, you have successfully:

- Created and connected a MongoDB Atlas cluster.
- Linked your database with MongoDB Compass.

- Performed CRUD operations in Node.js.

You're now ready to build backend systems that handle real-time data using MongoDB.